

Harshith Devaraja

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 LinkedIn —  GitHub —  Portfolio

Professional Summary

AI & ML Engineer with expertise in LLM evaluation, computer vision, NLP, and time-series forecasting. Experienced building scalable AI systems using FastAPI, Docker, MLflow, Redis, GitHub Actions, and GCP for deployment, automation, and real-time inference. Strong background in applied mathematics with hands-on experience in MLOps, model evaluation, reliability engineering, and distributed AI workflows.

Technical Skills

MLOps & Infrastructure: Docker, Kubernetes (GKE), FastAPI, MLflow, CI/CD, GitHub Actions, Celery, Redis, Streamlit, REST APIs

Cloud & Deployment: Google Cloud Platform (GCP), Cloud Run, Render, Supabase, Upstash Redis

Programming & Data: Python, SQL, R, Pandas, NumPy, PySpark

Machine Learning: Feature Engineering, Statistical Modeling, Drift Detection, Cross Validation, PCA, Clustering, Time-Series Forecasting

Deep Learning & NLP: PyTorch, TensorFlow, Transformers, PatchTST, FinBERT, CNNs, LSTMs, RLHF, SentenceTransformers

Mathematics: Linear Algebra, Statistics, Optimization, Haar Wavelets, Calculus

Professional Experience

Innodata Inc.

Oct 2025 – Present

AI Engineer / LLM Evaluation Engineer

- Engineered automated benchmarking pipelines in Python, reducing manual evaluation latency by **40%**.
- Designed quantitative evaluation frameworks for large language models, improving scoring consistency by **30%**.
- Performed hallucination analysis, anomaly detection, and reliability validation for production LLM systems.
- Collaborated with RLHF teams to integrate evaluation feedback into iterative training workflows.
- Technologies Used:** Python, RLHF, LLM Evaluation, Statistical Modeling, Prompt Engineering

Selected Projects

LLM Reliability & Evaluation Platform

 GitHub Live

- Built an AI reliability platform for monitoring and evaluating LLM outputs using FastAPI, Celery, Redis, and Streamlit.
- Implemented asynchronous evaluation pipelines and telemetry tracking for scalable model benchmarking and reliability analysis.
- Designed drift detection systems using Kolmogorov–Smirnov Test and Wasserstein Distance for monitoring model performance degradation.
- Deployed distributed AI services using Render, Streamlit Cloud, and Upstash Redis with cloud-native MLOps architecture.

Haar-CXR — Dual-Branch Medical AI System

 GitHub Live

- Developed a hybrid CNN architecture combining ResNet18 and Haar wavelet features for pneumonia detection from chest X-rays.
- Achieved **91% accuracy** and **0.94 ROC-AUC** using feature fusion and stratified cross-validation.
- Implemented explainable AI using Grad-CAM and uncertainty estimation with Monte Carlo Dropout.
- Built modular ML pipelines with MLflow, Docker, and Streamlit for experiment tracking and deployment.

AI-Powered SBI Financial Forecasting System

 GitHub Live

- Built a financial forecasting system using PatchTST Transformers and FinBERT sentiment analysis for SBI stock prediction.
- Engineered forecasting features including RSI, EMA, MACD, Bollinger Bands, volatility indicators, and lag-based signals.
- Achieved approximately **17% MAPE** and **212 RMSE** while generating AI-assisted BUY/SELL trading signals.
- Automated data ingestion, model inference, and alert pipelines using GitHub Actions, Supabase, and Telegram APIs.

Education

Manipal Academy of Higher Education

2022 – 2024

M.Sc. Applied Mathematics & Computing

University of Mangalore

2019 – 2022

B.Sc. Mathematics, Physics, Computer Science

Certifications

- IBM Generative AI Engineering
- IBM Machine Learning Specialization
- Machine Learning — University of Michigan
- Deep Learning Specialization